Duke University **October 21, 2021**
Math 105L: Laboratory Calculus and Functions I

Big Assignment 2

- This assignment covers work from Day 5-1 to Day 10-1.
- You have ten days to complete the assignment. It is due October 29 at 8am.
- You may use the worksheets, labs, and your notes, as well as any other resources.
- However, you may not work with anyone else on this assignment – it must be entirely your own work.
- Throughout the assignment, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers. For example, copying graphs from graphing software is insufficient 'work'.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.
- You must hand in a scanned version of this assignment on Gradescope and no other pages. If you need to do rough work, do it elsewhere before writing your answers.
- The number of points on each question is shown next to the question number. There are 132 points available in total, plus four bonus points.
- You must sign the statement at the bottom of this page.

Honor Code:

I have adhered to the Duke Community Standard and all the rules above in completing this assignment.

Signature: _____

1. (10 points) Solve the following equations. Write your answer in exact form, no decimals.

   (a) $10^{2x+3} = e^{x+7}$.

   (b) $\frac{1}{2}\ln(e^{2x^2-4x}) = -1$.

2. (13 points) Compute the derivative of $f(x) = \frac{1}{x^2}$ from the limit definition of the derivative (i.e. not using any of the differentiation rules we learned, such as the power, product, or quotient rules). You may use either the $\lim_{h \to 0}$ definition or the $\lim_{z \to x}$ definition.

3. (10 points) Suppose that the function $M(s)$ gives the number of hundreds of mosquitoes in a forest pond when the number of spiders present is s.

   (a) Explain the following mathematical statements in words:

      i. $M(20) = 10$.

      ii. $M'(20) = -0.5$.

   (b) Assuming that $M'(s) < 0$ for all sufficiently large values of s (say, $s > 10$ spiders), explain why it must be true that $M''(s) > 0$. Feel free to draw a picture supporting your answer.

4. (18 points)

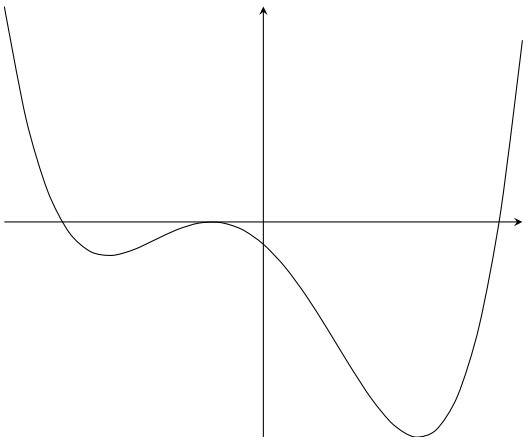
(a) Consider the function $f(x) = \frac{x-x^2-1}{e^x}$.

i. Show that $f'(x) = \frac{x^2-3x+2}{e^x}$.

ii. Compute the tangent line to $f(x)$ at $x = 0$.

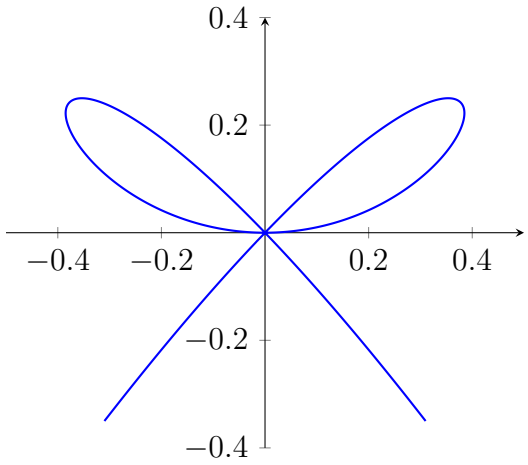
iii. For what values of x is $f(x)$ increasing?

(b) The derivative $g'(x)$ of a function $g(x)$ is shown below. Highlight on the graph all intervals in which ***the function*** $g(x)$ is concave down.



NOTE THAT THIS IS NOT THE GRAPH OF $g(x)$!

5. (27 points) Consider the curve defined implicitly by $x^4 = x^2y - y^3$. A picture of this curve (called the 'Bow Curve') is shown below.



(a) Show that $\frac{dy}{dx} = \frac{4x^3 - 2xy}{x^2 - 3y^2}$.

(b) Show that the curve has horizontal tangent lines when $y = \frac{1}{4}$ and vertical tangent lines when $y = \frac{2}{9}$. Draw these tangent lines on the curve.

(c) By observing the curve, state how many tangent lines it has at the point $(0, 0)$, and write down what one of them is.

6. (18 points) Suppose you invest $\$A$ in a bank account with annual interest rate r.

   (a) If the interest is compounded once a year, write down a formula for $O(t)$, the amount of money you have after t years:

   (b) If the interest is continuously, write down a formula for $C(t)$, the amount of money you have after t years:

   (c) Compute $O^{-1}(t)$ and $C^{-1}(t)$.

   (d) Use your answers from the previous part to compute the doubling time in each of the two cases if the interest rate is 6%. Express your answer in years and months, rounding to the nearest month.

7. (10 points) Suppose that $f(x)$ and $g(x)$ are differentiable functions.

(a) Show that

$$\frac{d}{dx}\left(\frac{f(x)}{[g(x)]^2}\right) = \frac{f'(x)g(x) - 2g'(x)f(x)}{[g(x)]^3}$$

(b) Use the result in (a) to find the derivative of $h(x) = \frac{2^x}{(\ln x)^2}$.

8. (18 points) Consider the piecewise function

$$f(x) = \begin{cases} ae^{2x} & -\infty < x \leq 0 \\ x^2 + (b+3)x + 2b & 0 < x < \infty \end{cases}$$

(a) Show that this function is continuous if $a = 2b$.

(b) Find values of a and b that make $f(x)$ differentiable and find the value of $f'(x)$ at $x = 0$.

9. (8 points) Each of the following statements is *false in general* (though may be true for particular instances). For each, briefly explain why it is false in general.

   (a) If $f(x)$ is concave up, then $f''(x) > 0$.

   (b) We can always find $f^{-1}(x)$ by swapping x and y and solving for y.

   (c) An increasing, concave down function must have a horizontal asymptote. (Hint: you know a function for which this isn't true! What is it?)

   (d) If $f(x)$ is continuous at $x = 0$, it is differentiable there.

Bonus Question (2 points) There are municipal elections coming up in Durham. Research the web presence of the two Ward 3 candidates, Leonardo Williams and AJ Williams. Write down two substantial policy differences between the two candidates.

Bonus Question (2 points) Draw a funny picture, paste in a good meme, tell me a good story, or tell me a good joke.